

CSE110A W26: HMWK5 Plus Extra Credit Ideas

CSE110A: Introduction to Compiler Design

HWK5: Covers the topic of Optimization involving Local Value Numbers, and Loop Unrolling. You have one week to complete it. It is worth 10% of your grade. You will have one week to do it.

HWK6: This one is Extra Credit: It is essentially a redo of HWK4 except you will be generating valid LLVM IR Code which then can become an executable that can work on your computer. You will either need to use the docker file provided or install LLVM for your computer to run it. Completing this can add an extra 5% to your grade.

Additional Extra Credit: Do a minimum of 3 pages summary about any of the videos or topic write-ups below. You can use ChatGPT, but you need to include the questions and answer, and at least one paragraph of how you used it and why.

Additional Extra Credit: Giving a 5 minute talk (plus) on the last day of class (Thursday 3/12) on a topic you choose.

The grading details of the extra credit, are still being worked out. For the present think of it as additional learning experience.

The topics below are suggestions: A good source of interesting topics about compilers is the LLVM Channel in YouTube. In a way the farther back you go, the less esoteric the topics are. LLVM IR, MLIR, and tool chain related to LLVM are all great topics. Topics related to Flex, Lex, Yacc, Bison, etc.

You can also do something interesting in the LLVM pipeline. e.g. something cutting edge provocative that could stimulate future work for yourself and others. e.g. suggest a novel idea. You can do this yourself and/or collaborate with a 2nd person.

Possible things to cover:

- Make a critique of the video, from a learning perspective
- Interesting aspects of what you saw.
- Mini-tutorial of what you learned.
- Review of novelty of ideas you saw.
- Potential for future research.
- Recommendations.

- Make it interesting to yourself and others.

Publications and Tutorials

A Gentle Introduction to LLVM IR:

Website with a tutorial

<https://mcyoung.xyz/2023/08/01/llvm-ir/>

SITE: LLVM

2024 EuroLLVM - Teaching MLIR concepts to undergraduate students

5.2K views • 11 months ago 19:01

2019 EuroLLVM Developers' Meeting: V. Bridgers & F. Piovezan LLVM IR Tutorial - Phis, GEPs ...

29K views • 6 years ago 50:46

2017 EuroLLVM Developers' Meeting: D. Majnemer "XLA: Accelerated Linear Algebra"

5.4K views • 8 years ago 36:41

2024 EuroLLVM - MLIR Linalg Op Fusion - Theory & Practice

1.5K views • 11 months ago 21:13

2019 LLVM Developers' Meeting: A. Warzynski "Writing an LLVM Pass: 101"

16K views • 5 years ago 1:09:15

2017 LLVM Developers' Meeting: M. Braun "Welcome to the back-end: The LLVM machine representation"

9.8K views • 7 years ago 53:56

2020 LLVM in HPC Workshop: Keynote: MLIR: an Agile Infrastructure for Building a Compiler Ecosystem

9.6K views • 4 years ago 1:04:43

SITE: LLVM Social Bangalore

Deep Learning Based Compiler Optimizations

1.9K views • 3 years ago 1:42:38

Introduction to Compilers and Stages of Compilation

3.1K views • 2 years ago 1:11:37

Torch-MLIR 101

3K views • 2 years ago 45:00

SITE: Triton

Mosaic GPU: A DSL for Fast Hopper Kernels in Python

801 views • 8 months ago 15:59

Compiler Tools: Writing an MLIR Pass

2K views • 8 months ago 31:51

SITE: TensorFlow

Inside TensorFlow: AutoGraph

10K views • 5 years ago, 53:55

Inside TensorFlow: TF Model Optimization Toolkit (Quantization and Pruning)

16K views • 5 years ago 42:35

Inside TensorFlow: MLIR for TF developers

10K views • 5 years ago 43:41

TOPIC: SPARSE TENSORS AND COMPILERS

[CTSTA'23] Challenges and Opportunities for Sparse Compilers in LLM

19 views • 10 months ago 16:27

ACM SIGPLAN

ICPP Keynote: Making Sparse Fast with the TACO Domain Specific Language for Sparse Tensor Algebra

194 views • 4 years ago 30:12

Saman Amarasinghe

AMD HIP Tutorial, 9-3, rocSPARSE

117 views • 1 year ago 13:54

Yifan&GPU

[SPARSE24] Design DSLs with xDSL

240 views • 10 months ago 21:33

ACM SIGPLAN

MISCELLANEOUS SITES:

How to build a compiler with LLVM and MLIR - 09 IR (SLIR) generation

2.6K views • 3 years ago 1:04:45

lxsameer

Read a paper: Multi-level Intermediate Representation (MLIR)

6.9K views • 3 years ago 7:42

Vivek Haldar

Who uses JAX?

10K views • 2 years ago 3:31

Google Research

Building domain-specific compilers quickly with MLIR compiler infrastructure |

Chris Lattner

16K views • 4 years ago 4:30

Lex Clips

HC34-T2: Heterogeneous Compilation in MLIR

5.5K views • 2 years ago 3:32:32

hotchipsvideos

Writing a compiler with LLVM - Cailin Smith - NDC Oslo 2022

48K views • 2 years ago 47:58

NDC Conferences

What's New in Compiler Explorer? by Matt Godbolt

https://www.youtube.com/watch?v=Ey0H79z_pco&t=6s

2.3K views 1 year ago #cpp #cppprogramming #compiler

<https://www.cppnorth.ca>

An introduction to LLVM IR

<https://www.youtube.com/watch?v=CDKuH7SIgdM&t=614s>

2.2K views 3 months ago 33:55